MOHAMMAD AMANOUR RAHMAN

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SUMMARY

Researcher focused on building reliable and interpretable deep learning systems for medical imaging, with a self-directed research program spanning image segmentation, explainable AI, and uncertainty estimation. Author of multiple peer-reviewed journal and conference publications, primarily as sole or lead researcher. Motivated by developing AI systems that are accurate, transparent, and clinically trustworthy, seeking to continue this research path through a PhD under close academic mentorship.

RESEARCH INTERESTS

Medical Vision & Representation Learning · Trustworthy AI & Uncertainty · Multimodal & Federated Learning

Undergraduate Thesis: Automated Brain Hemorrhage Detection, Segmentation, and Severity Quantification from Non-Contrast CT Using a Hybrid Deep Learning Framework — supervisor: Dr. Kazi A Kalpoma, Professor, CSE, AUST

EDUCATION

Ahsanullah University of Science and Technology (AUST) **Dhaka, Bangladesh**
B.Sc. in Computer Science and Engineering *2022 – 2026*

PUBLICATIONS

Conference & Workshop Papers

1. Rahman, M.A. "UCBound-Net: Uncertainty-Guided Boundary-Aware Continual Learning for Domain-Incremental Ultrasound Segmentation." MICCAI 2026 CLiMeM Workshop. Accepted. DOI: 10.48550/arXiv.2608.01518
2. Rahman, M.A. "MedSaab-US: A Backpropagation-Free Multi-Scale Wavelet-Saab Framework for Thyroid Nodule Segmentation in Ultrasound Images." IEEE ICIP 2026 Satellite Workshop. Accepted. DOI: 10.48550/arXiv.2607.02209
3. Rahman, M.A. "RadiomicNet: A Hybrid Radiomics-Guided Lightweight Architecture for Interpretable Medical Image Segmentation." IEEE ICIP 2026 Satellite Workshop. Accepted. DOI: 10.48550/arXiv.2607.02185

Journal Articles

Published

4. Rahman, M.A. "TCB-Net: Topology-Constrained Boundary-Interior Decoupled Network for Breast Ultrasound Lesion Segmentation." IET Image Processing (Wiley), 2026. DOI: 10.1049/ipr2.70445
5. Rahman, M.A., Prerona, R.Z. "Spectral-Spatial Mamba with Uncertainty-Guided Refinement for Thyroid Nodule Diagnosis." Array (Elsevier), 2026. DOI: 10.1016/j.array.2026.101058
6. Rahman, M.A. "HyFormer-Net: A Synergistic CNN–Transformer with Interpretable Multi-Scale Fusion for Breast Lesion Segmentation and Classification in Ultrasound Images." Intelligence-Based Medicine (Elsevier), 2026. DOI: 10.1016/j.ibmed.2026.100413

Under Review

7. Rahman, M.A. "Joint Thyroid Nodule Segmentation and TI-RADS-Aligned Boundary Morphology Typing via a Dual-Head Network Trained on Automatically Generated Pixel-Level Labels." Scientific Reports. Under Review.
8. Rahman, M.A. "TIGRA: TI-RADS-Informed Graph Reasoning Architecture for Explainable Thyroid Nodule Segmentation in Ultrasound Images." Information Sciences. Under Review.
9. Rahman, M.A. "FedXAI: Federated Medical Imaging with Intrinsic Explainability and Differentially Private Explanation Auditing" Machine Learning. Under Review.
10. Rahman, M.A. "OADCAI: Ordinal-Aware Multimodal Deep Learning for Three-Class Pulmonary Nodule Classification Including Indeterminate Cases." PLOS ONE. Under Review.

Poster

Automated Brain Hemorrhage Detection, Segmentation, and Severity Quantification from Non-Contrast CT Using a Hybrid Deep Learning Framework — Presented as a poster at AUST.

RESEARCH EXPERIENCE

Independent Researcher — Medical Image Analysis & Deep Learning **2024 – Present**

- Developed CNN–Transformer (HyFormer-Net), topology-constrained (TCB-Net), and spectral-spatial Mamba (SSMURNet) architectures for breast and thyroid ultrasound lesion segmentation and classification.

- Designed a backpropagation-free wavelet-Saab framework (MedSaab-US) and a radiomics-guided lightweight architecture (RadiomicNet) for interpretable segmentation, and an ordinal-aware multimodal classifier (OADCAI) for CT-based pulmonary nodule diagnosis.

Lead Researcher, Undergraduate Thesis **2025 – 2026**

- Automated Brain Hemorrhage Detection, Segmentation, and Severity Quantification from Non-Contrast CT Using a Hybrid Deep Learning Framework.
- Developing the framework and a Multi-Dimensional Imaging Severity Index (MDISI) for clinical-facing severity scoring, in collaboration with a neurosurgeon for validation; intended for submission to a leading medical image analysis journal.

Earlier Projects

EfficientNet-CBAM for Brain Tumor Classification *Github*

- Developed an EfficientNetB3-CBAM framework for multi-class brain tumor MRI classification, achieving 99.29% accuracy; integrated HiResCAM for explainability. Tech: TensorFlow, Keras, OpenCV.

Deforestation Mapping in Dhaka — Semantic Segmentation *Github*

- Built a high-resolution satellite imagery dataset (2012–2022) and benchmarked U-Net variants (Dice, IoU, F1, Precision, Recall) for deforestation mapping. Tech: Python, TensorFlow, PyTorch, OpenCV.

ACADEMIC SERVICE

Peer Reviewer

- MICCAI 2026 Efficient Medical AI (EMA4MICCAI) Workshop, — **Best Reviewer Award**
- Med-Reasoner 2026 NeurIPS Workshop
- PLOS ONE

Open-Source Contributor — Project MONAI (NVIDIA & King's College London)

- Implemented 5-Fold Cross-Validation for UNETR (PR #422, MONAI Research-Contributions): added a `--fold` argument and integrated MONAI's `partition_dataset` API for fold-indexed training/validation.
- Resolved High-Version MONAI Compatibility Issues (PR #421): fixed a crash from the deprecated `pos_embed` parameter in ViT under MONAI 1.3.0+, using a version-aware runtime compatibility mechanism built with Python's `inspect` module.
- Resolved a documentation defect in the PerceptualLoss module (Issue #8592), improving usability and reproducibility for the broader community.

RESEARCH SOFTWARE

MedXAI — Open-Source Python Toolkit for Medical Imaging AI

PyPI | GitHub

- Independently developed and maintains MedXAI, a lightweight open-source toolkit for reproducible medical imaging AI research, distributed via PyPI and GitHub.
- Provides modular utilities for segmentation metrics, custom loss functions, image preprocessing, radiomics feature extraction, and explainability (Grad-CAM) for PyTorch-based research workflows.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, MATLAB

Deep Learning Frameworks: PyTorch, TensorFlow, Keras, MONAI

Computer Vision & Image Processing: OpenCV, scikit-image, Albumentations, Pillow

Scientific Computing & Data Analysis: NumPy, SciPy, Pandas, Matplotlib

Research & Software Development: Git, GitHub, GitHub Actions, PyPI, Jupyter Notebook, Google Colab, VS Code

REFERENCES

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